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Secure IOT Framework: Blockchain Authentication and Intrusion Detection

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Abstract

Blockchain security leverages decentralized, immutable ledger technology for identity verification and data integrity but faces challenges like high computational overhead, scalability issues, and privacy-explainability trade-offs. To overcome these, this research proposes a framework integrating the Variable Elliptical Anonymous Candidate protocol (VEAQS) with blockchain for secure authentication, ensuring transparency and efficiency. The Quantum Kernel Convoluted Gazelle Self-attention model (QKCGS) enhances intrusion detection by leveraging quantum kernels and self-attention mechanisms. It efficiently handles sequential and contextual data, improving both short- and long-term dependency learning. The model uses quantum gates to regulate past knowledge retention and irrelevant data filtering, boosting detection accuracy. By integrating a deep learning model with SHAP, the framework enhances the interpretability and explains the ability of intrusion detection deployed in the Internet of Things, improving the detection of complex attack patterns while ensuring transparent decision-making. The model further optimizes authentication processes by minimizing computational overhead and latency through Single Candidate Optimization (SCO), streamlining the key generation process. The Gazelle Optimization Algorithm (GOA) is used to fine-tune the intrusion detection systems, making it more responsive to real-time attacks while maintaining the integrity and security of the authentication process. This combination of deep learning, blockchain security, and explainable Artificial Intelligence (AI) creates a robust, efficient, and transparent security solution for modern Internet of

Things systems, enabling secure communication, tamper-proof logging, and real-time attack detection. The proposed methods achieve 99.50% of accuracy, 99.52% precision, and 99.60% F1-score, demonstrating significant improvements over existing methods.

Keywords: Attack Classification, Authentication, Blockchain, Deep Learning, Intrusion Detection.

1. Introduction

Blockchain is a secure, decentralized digital ledger that stores records transparently, immutably, and tamper-resistant across a network of computers [1]. The merging of blockchain and AI by Dr Turing in 1950 has transformed business with data immutability, transparency, and efficiency [2]. Advances in blockchain technology with smart contracts and AI have improved scalability, with platforms like Ethereum and Avalanche offering solutions for private/public sub-networks [3]. However, integration of AI with blockchain raises challenges, with issues of scalability, regulatory oversight, and privacy concerns. The complexity and high costs associated with implementing blockchain technology also hinder its widespread adoption [4]. Despite these challenges, continuous technological advancements have the potential to revolutionize various industries in the future [5]. Blockchain is revolutionizing the healthcare system by increasing patient information accessibility and security through its decentralized principles [6]. This overturns the traditional healthcare hierarchy, allowing patients to govern their treatment. Healthcare has transitioned from paper-based systems to electronic records, paving the way for developments like telemedicine and IoT (Internet of Things) [7]. Blockchain potential impact on healthcare is vast, spanning from supply chain management to electronic health records and data analysis. Access control uses identity, role, or attribute methods. [8]. Attribute-Based Signature (ABS) is proposed for signer anonymity due to the inadequacy of public-key encryption. Protocol merges ABE, ABS with trust-based access control using ledger fabric [9]. Smart cities leverage IoT technology to tackle urbanization challenges, prioritizing factors like centralization, scalability, trust, privacy, and security [10]. Hyper ledger Fabric blockchain in IoT raises privacy concerns, highlighting the uniqueness of the proposed computational trust value application [11]. The approach analyzes parameters like user behavior, attributes, trust levels, unauthorized requests, and policy requirements [12].Blockchain technology shows promise in healthcare, include patient data management, drug traceability, and data security, with various blockchain types and consensus mechanisms

explored [13]. Despite its security benefits, challenges remain in leveraging blockchain effectively in healthcare, requiring ongoing research and solutions. Knowledge-Defined Networking (KDN) integration, handling big data challenges, can be improved using techniques like sharing, compression, fragmentation, parallel processing, and off-chain storage [14]. Interoperability between KDN systems using different blockchain frameworks can be enhanced with interoperability frameworks like PIE Chain and Interchain [15]. Building blockchain networks with directed acyclic structures can increase capacity and parallel processing capabilities [16]. Lightweight and low-energy-consuming blockchain frameworks like Sensor-Chain can improve scalability. Sharding, pruning, and high-throughput blockchain frameworks like Conflux can reduce latency and enhance performance [17]. Security vulnerabilities can be addressed using Machine Language (ML) algorithms, encryption, and AI experts.

Motivation for this research arises from the potential of integrating blockchain and AI to revolutionize industries such as healthcare and smart cities by enhancing transparency, security, and efficiency. However, challenges such as scalability, privacy concerns, and high implementation costs hinder adoption. The complexity of combining these technologies, along with interoperability and energy efficiency issues, complicates their integration. Addressing data sharing, security, and decision-making in decentralized systems is essential. Optimizing blockchain frameworks and developing Explainable AI are crucial to overcoming these barriers. This research aims to provide solutions to unlock the full potential of AI-blockchain integration while enhancing security and authentication mechanisms. Major contributions of the research can be,

- The primary objective of this research is to develop a secure and decentralized authentication framework for IoT, leveraging blockchain technology and AI to ensure data integrity, trust, and efficiency in communication among connected entities.
- To achieve this, we propose the VEAQS framework, which integrates the Variable Elliptical Anonymous Physical Candidate (VEAC) protocol and the QKCGS model.
- The VEAC protocol enhances authentication by enabling secure and anonymous communication between IoT devices, reducing dependency on centralized authorities. Additionally, employing Physical Unclonable Functions (PUF) techniques minimizes key storage requirements while improving authentication security, ultimately reducing execution time and memory consumption.
- The QKCGS model, integrated with SHAP, enhances intrusion detection and classification by combining quantum computing with explainable AI.

- SCO is employed to reduce computational overhead and authentication latency by optimizing key generation. The GOA refines intrusion detection by efficiently searching and refining solutions, ensuring robust security mechanisms.

The remainder of the document is structured as follows: The review of the literature covered in Section 2. The proposed system description is presented in Section 3 and Section 4 covers results and discussion. The paper reaches its conclusion in Section 5.

2. Literature Survey

The literature review highlights major works on AI-blockchain integration for data protection and privacy.

In 2023, T.A. Shaikh *et al*. [18] have introduced Health Cyber-Physical Systems (HCPS) that would replace traditional medical equipment, operating with sensors and systems to monitor patients continuously. Dispersed frameworks had a successful history for MD-CPS development.

In 2024, M. K. Aiden et al. [19] have proposed blockchain technology and AI to enhance the security of CPS. As CPS becomes more interconnected with internet networks, vulnerabilities arise, and blockchain decentralized database system can address issues related to data storage, authentication, and interchange. AI can further analyze the collective data to predict and improve security measures.

In 2022, R.F. Mansour, *et al*. [20] have suggested advancements in AI to create effective intrusion detection models for CPS. Metaheuristic algorithms help reduce dimensionality.

In 2023, Sattar *et al.* [21] have presented a secure framework for online exams in Bangladesh that aims to improve efficiency by addressing identity verification and unauthorized access issues for students and teachers.

In 2022, P. Kumar, *et al.* [22] have introduced traditional security solutions like Intrusion Detection and Prevention Systems, which were often implemented after assets were operational, giving attackers detailed knowledge to launch attacks on vulnerable infrastructure, posing threats to public safety in industrial settings.

In 2024, S. Saifullah *et al.* [23] have proposed large amounts of data across multiple locations, which can lead to the leakage of sensitive information by neural networks. Lack of explain ability and privacy hinders the practical application in critical domains.

In 2024, S. Baccouri *et al.* [24] have introduced new Elliptic Curve ElGamal encryption method enhances speed and, omits lookup tables. Parties authenticate and exchange

parameters simultaneously for enhanced security. The scyther tool validates superiority over RSA (Rivest Shamir Adelman) in security and performance. Recent works comparison is tabulated in Table1.

Table 1: Table of comparisons for recent works

Authors	Focus	Technique	Advantages	Limitations
Shaikh *et al.* [18]	Smart healthcare, CPS	Deep Belief Network (DBN)	Enhances healthcare security	High computational power required for AI models; potential latency in real-time applications
Aiden *et al.* [19]	Cyber security in CPS	AI, Block chain	Combines AI with block chain for CPS security	Computational overhead in AI-based cyber security; latency issues in block chain-based authentication
Mansour *et al.* [20]	Intrusion detection in CPS	Cyber Physical System (CPS)	Optimizes security with AI	Focused on intrusion detection; scalability issues
Sattar *et al.* [21]	Block chain in online exams	Blockchain	Secures online exams	Limited to online exam use cases; not applicable to broader block chain security contexts
Kumar *et al.* [22]	Secure industrial IoT	Long Short-Term Memory (LSTM)	Enhances security in industrial networks	Scalability low and energy efficiency in large-scale industrial IoT networks
Saifullah *et al.* [23]	Privacy-explain ability in federated learning	Black-box Neural Network (BNN)	Improves privacy and security	Complexity in managing the privacy-explainability

Baccouri *et al.* [24]	Lightweight authentication	Wireless Sensor Networks (WSN)	Efficient cryptographic authentication	Limited to elliptic curve cryptography; may not handle advanced attacks

2.1. Problem Statement

Blockchain security systems face multiple challenges, particularly high computational overhead in key storage, inefficient cryptographic authentication mechanisms, and the lack of explainability in AI-based intrusion detection. These limitations significantly hinder real-time performance in both authentication and intrusion detection tasks. As blockchain networks scale, transaction processing efficiency becomes a bottleneck, complicating the ability to handle high-volume transactions with low latency. Challenges in key generation lead to significant computational overhead and latency, impacting system efficiency and scalability. Existing authentication methods, such as RSA and ECC (Elliptic Curve Cryptography), rely on centralized key storage, making them vulnerable to security breaches and computationally expensive in large-scale environments. PUF has been underused in blockchain authentication due to limited integration with cryptographic techniques. It is utilized to generate challenge-response pairs for device authentication, eliminating centralized key storage and reducing overhead. Current intrusion detection solutions, including deep learning-based AI models, often function as black-box systems, limiting transparency and explainability. These models struggle to balance detection accuracy with privacy concerns, as adversarial attacks can exploit AI weaknesses. Additionally, traditional cryptographic techniques, such as AES (Advanced Encryption Standard) and SHA-based hashing, face challenges in adapting to evolving cyber threats. To systematically address these challenges, this research proposes the VEAQS framework, which enhances blockchain authentication through Secure Cryptographic Operations, optimizing key generation while reducing computational overhead. The model integrates(ECDS) and PUF to improve security without the need for centralized key storage. For intrusion detection, quantum kernels and self-attention mechanisms are incorporated to enhance attack pattern recognition and improve detection accuracy. Additionally, the GOA refines feature selection by optimizing exploration and exploitation within the search space, leading to faster and more precise threat detection.

3. Proposed Methodology

The goal of this research is to enhance security, interpretability, and intrusion detection in IoT networks using a blockchain-based authentication mechanism combined with AI-driven attack detection. The input data is collected from the CIC-IDS-2017 and CSE-CIC-IDS-2018 datasets, passing through multiple security layers, starting from the consumer IoT device to the IoT-to-Fog gateway, and then to the Fog server. To ensure tamper-proof authentication, the data first undergoes Genesis Block Mining, which establishes a secure and immutable foundation in the blockchain. This process verifies transactions, preventing unauthorized modifications and ensuring trust and transparency before the data reaches the Cloud server. Once authenticated, the data is processed by the QKCGS for intrusion detection, efficiently classifying attacks. Additionally, SHAP is utilized to enhance model interpretability, ensuring accurate attack classification. This integrated approach addresses key challenges such as scalability, computational efficiency, and privacy-preserving explainability, making IoT environments more secure and resilient against cyber threats. The schematic representation of the VEAQS proposed model is shown in Figure 1.

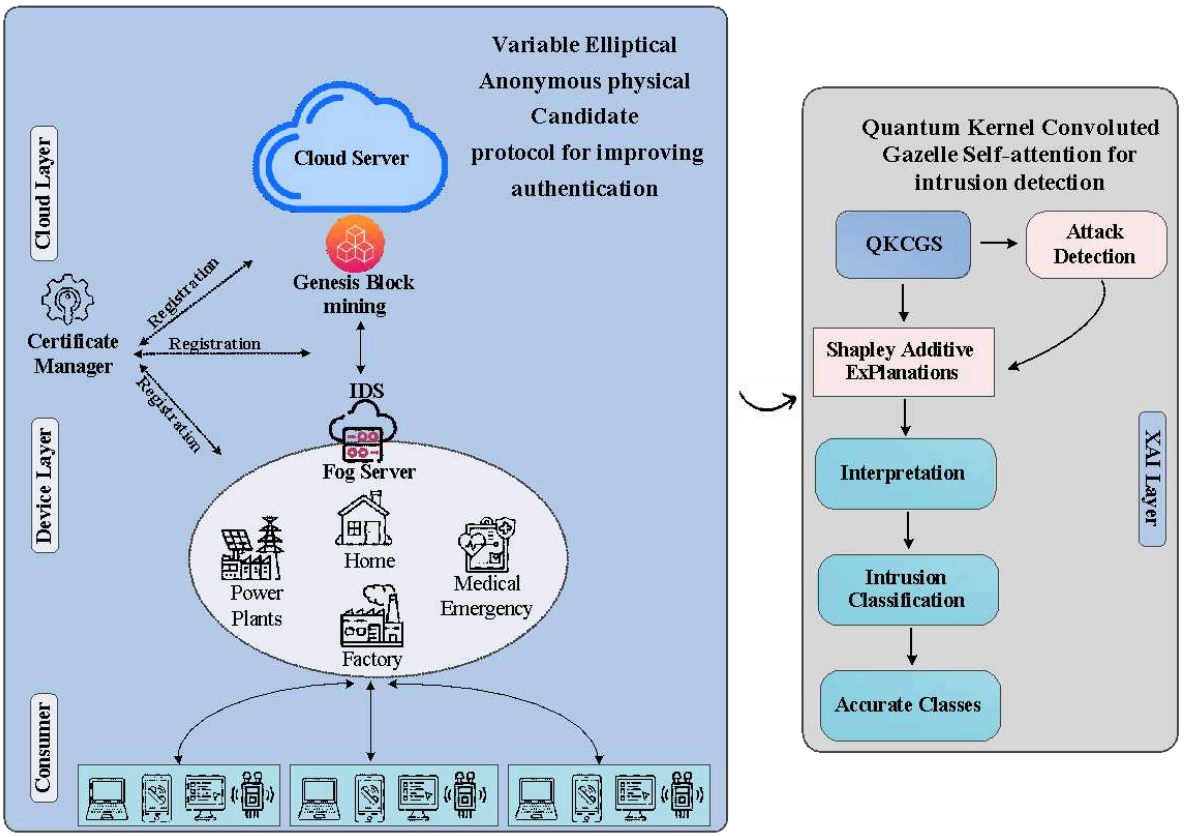


Figure 1: Schematic Representation of VEAQS proposed model

3.1. Variable Elliptical Anonymous physical Candidate (VEAC) protocol for improving authentication

The VEAC protocol [25] enhances authentication security in IoT environments by enabling secure, anonymous communication between IoT devices and fog nodes. In the IoT-to-IoT authentication phase, IoT devices (n1 and n2) use the VEAC protocol to establish secure communication. IoT n1 generates a signature with its device ID, timestamp, and secret values, which IoT n2 verifies. If the signature is valid, IoT n2 generates a session key and sends it back with another signature for IoT n1 to verify. In the IoT-to-Fog phase, IoT n and the Fog node (FGN) follow a similar process to generate a shared session key after mutual verification. In the Fog-to-Cloud phase, the Fog node and Cloud Service Provider (CSP) also verify each other's signatures and timestamps to establish a secure key. The Genesis block serves as the foundation of the blockchain, containing essential metadata such as a unique identifier, timestamp, and cryptographic signatures to ensure authenticity and integrity. This block is hashed and signed using a private key, providing a reference point for all subsequent blocks. The hash of the Genesis block is used in the next block's formation, maintaining the blockchain's security and immutability. Unlike regular blocks, the Genesis block does not store transaction data; instead, it acts as a starting reference for the blockchain, ensuring its structural integrity. Its data is excluded from further calculations, serving solely as an initial anchor for the chain. PUF [26] generates unique challenge-response pairs for device authentication and secure authentication without the need for storing secret keys, while ECDS is used for secure signing and verification of messages, leveraging elliptic curve techniques within block chain systems. ECDS and PUF are integrated for authentication, as shown in equations (1) and (2).

$$z^2 = y^3 + vy + w(\text{mod } r) \tag{1}$$

$$y \cdot Q = Q + Q + \cdot + Q \ (y \ \ times) \tag{2}$$

where, $y \, and \, z$ represents the coordinates point on the elliptic curve, v, r represents the Coefficients that define the specific elliptic curve, Q represents the generated public keys in an elliptic curve and $y \cdot Q$ represents the scalar multiplication of the point Q. The Output for PUF is shown in (3) and (4),

$$y \cdot Q + z \cdot Q = (y + z) \cdot Q \tag{3}$$

$$S_j = PUF(D_j) \tag{4}$$

where, S_j represent PUF response output and D_j represent PUF challenge input that influences computational overhead and latency during authentication. Once the validation process is completed, encryption begins to secure the block before it is registered on the blockchain. The important information, including the index, user ID, data, previous block's hash, and timestamp, is first hashed using a cryptographic hash function. This step ensures that the data is condensed into a fixed-size string, representing the block's contents uniquely. Next, the hashed information is signed using the user's private key, providing authentication and ensuring that the block has not been tampered with. The final block now includes the index, hash, signature, timestamp, and user ID. This encrypted data is then ready for secure registration on the block chain. The encryption and signing process guarantees that the block's contents are both confidential and immutable, ensuring trust and security in the system. In ECDS, multiple prime keys are generated to ensure strong security, but this results in significant computational overhead and latency during the authentication process. The large number of keys and complex calculations required for key generation can slow down system performance.

3.1.1. Single Candidate Optimization (SCO)

It [27] is used to optimize the key generation process by selecting a smaller, more efficient set of candidate primes. This reduces the computational burden and minimizes latency during authentication.

Fitness Function:

Fitness Function of te

$$Fitness\ Function = \min(D_j) \tag{5}$$

where, D_j represents the minimizing of computational overhead and latency during authentication.

Exploration phase:

Update the candidate solution position in equation (6),

$$y_k = \begin{cases} h\,best_k + (x \mid h\,best_k \mid) & if\ \ s_1 < 0.5 \\ h\,best_k - (x \mid h\,best_k \mid) & otherwise \end{cases} \tag{6}$$

where, s_1 represents the range of random variables from [0,1] and the mathematical definition for the x can be shown in equation (7),

$$x(u) = \exp^{-\left(\frac{cu}{U}\right)^c} \tag{7}$$

where, c is a constant, u represents the function evaluation of current iteration and U represents the most function evaluations possible.

Exploitation Phase:

In phase two, the search focuses on refining the best solution identified in phase one, where the candidate solution updates its position based on the modifications made during this phase as shown in (8).

$$y_k = \begin{cases} hbest_k - ((s_2 x(vc_k - mc_k))) & if\ s_2 < 0.5 \\ hbest_k - ((s_2 x(vc_k - mc_k))) & otherwise \end{cases} \tag{8}$$

where, s_2 represents the another range of random variables from [0,1], $hbest_k$ represents the global best position k, vc_k represents the upper k bound and mc_k denotes lower k bound. In the second phase of SCO, potential solution adjusts position according to equation (9).

$$y_k = \begin{cases} hbest_k + ((s_3 (vc_k - mc_k))) & if\ s_3 < 0.5 \\ hbest_k - ((s_2 (vc_k - mc_k))) & otherwise \end{cases} \tag{9}$$

where, s_3 represents random variables from [0,1] range. When the positions of certain variables are updated, their values may occasionally exceed the specified range or boundaries. To prevent this, the updated positions are adjusted to ensure they do not surpass the upper or lower bounds. If a value exceeds the upper bound, it is set to the bound and similarly for the lower bound is shown in (10).

$$y_k = \begin{cases} hbest_k & if\ y_k > vc_k \\ hbest_k & f\ y_k > mc_k \end{cases}$$

(10)

The initial solution is generated and shown in (11),

$$y_k = mc_k + s_4 (vc_k - mc_k)$$

(11)

Thus, the minimizing of computational overhead and latency during authentication is essential for improving system efficiency. SCO achieves this by streamlining the authentication process, reducing resource consumption. As a result, it enables faster, more efficient the authentication without compromising security. After finishing authentication

process the data comes to the QKCGS for detection of intrusion. Pseudocode for Genesis Block Mining is shown in Table 2.

Table 2: Pseudocode for Genesis Block Mining

```
class GenesisBlock:
      def __init__(self, index, previous_hash, user_id, private_key, puf_response):
       self.index = index
       self.timestamp = str(time.time())
       self.previous_hash = previous_hash
       self.user_id = user_id
       self.puf_response = puf_response
       self.hash = self.generate_hash()
       self.signature = self.sign_block(private_key)
  def generate_hash(self):
    return hashlib
  def sign_block(self, private_key):
    return private_key + self.hash
  def mine_block(self):
     while not self.hash.startswith("0000"):
       self.timestamp = str(time.time())
       self.hash = self.generate_hash()
       self.signature = self.sign_block(private_key)
   return self
```

The Genesis Block Mining process ensures secure blockchain authentication. It initializes the Genesis Block with attributes like index, timestamp, previous hash, user ID, and PUF response to enhance security. The generate_hash () function creates a unique cryptographic hash, while sign_block () ensures data integrity using a digital signature. The mining process updates the block's hash iteratively until it meets a predefined difficulty criterion, ensuring immutability. By integrating PUF-based authentication with ECDS, this approach enhances security while reducing computational overhead.

3.2. Quantum Kernel Convoluted Gazelle Self-attention (QKCGS) for intrusion detection

QKCGS [28] leverages the power of quantum kernels and self-attention mechanisms to enhance the detection of complex attack patterns. It is designed to handle sequential and

contextual data more effectively, improving both short- and long-term dependency learning in intrusion detection.

3.2.1. Self-attention-based quantum kernel mechanism

The model uses two gates to control information flow: one gate regulates the count of past knowledge retained, while the other determines the amount of irrelevant data to discard. In sequential data, it may effectively manage both short-term and long-term dependencies. It increases the precision of intrusion detection in dynamic contexts by processing data both forward and backward.

$$QKSAM := |W\rangle \Theta(\theta_4)_{|\langle Q|K\rangle|^{2-}} \tag{12}$$

Equation (12) highly and abstractly recapitulates the below concepts, encompassing Quantum Query State (QQS) $|R\rangle$, Quantum Valued State $(QVS)|V\rangle$, Quantum Key State $(QKS)|K\rangle$ Quantum Kernel Self-Attention Score $(QKSAS)|\langle Q|K\rangle|2$, and mathematical connections $\Theta(\theta 4)$ between quantum registers. The self-attention score can be shown in (13),

$$x_j = [b_d]_{d=0}^{m-1}, x_k = [c_d]_{d=0}^{m-1} \in S^{1\times m} \subseteq Y \tag{13}$$

where, b_d and c_d are elements in x_j and x_k, m represents the dimension of the element and Y represents the spatial domain where the input data resides.

$$V_\phi : x_j \to \sum_{d=0}^{m-1} \alpha_d |c\rangle_1 + \sum_{d=m}^{2n-1} 0|d\rangle_1 \subseteq I \tag{14}$$

$$V_\phi : x_j^u \to \prod_{d=0}^{m-1} \left(\bigotimes_{e=0}^{m-1} S_f \left(b_{e+d\times m}^1 \right) \right) |0\rangle_1 \subseteq I \tag{15}$$

$$|R\rangle = V(\theta_1) V_\phi (xj) |0\rangle_1 \subseteq I$$

(16)

$$< L | = < 0 |_1 V_0^\dagger (x_k) V^\dagger (\theta_2) \subseteq I^\dagger \tag{17}$$

$$|< R | L >|^2 = |< 0 |_1 V_\phi^\dagger (x_k) V^\dagger (\theta_2) V(\theta_1) V_\phi (x_j) | 0 >_1|^2 \tag{18}$$

where, $R\rangle$ represents the structure of the parameter layers $V(\theta_1)$ and $|< R | L >|^2$ represents the quantum kernel similarity measure.

3.2.2. Multi-Class Classification with QKCGS

This approach uses a multi-class classification strategy to categorize different types of attacks. The proposed IDS incorporate algorithms like Softmax to compute the probability distribution across multiple attack categories. The attack types are Benign, Dos Hulk, Bot, Comb, DDOS attack-HOIC, DDoS (Distributed Denial of Service), DDoS attacks-LOIC-HTTP, Port Scan, DoS Golden Eye, FTP-Patator, DoS attacks-Slow HTTP Test, Web Attack SQL Injection, DoS attacks-Slowloris, Heartbleed, FTP-Brute Force, SSH-Patator, Infiltration, Heartbleed, SSH-Brute Force, Web Attack Brute Force, and Web Attack XSS. This enables the model to accurately classify attacks into the correct class based on the input data and predict the most probable attack type.

3.2.3. Gazelle Optimization Algorithm (GOA) for minimizing error

GOA [29] is chosen for its population-based approach, which effectively explores and exploits the search space through diverse candidate solutions. It optimizes the fitness function by minimizing kernel errors and improving attack detection accuracy through efficient search space traversal.

Population Initialization

GOA is an optimization algorithm based on population, where randomly initialized gazelles (A) serve as search agents represented by $n \times x$ matrix of potential solutions can be shown in (19),

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,x-1} & a_{1,x} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,x-1} & a_{2,x} \\ \vdots & \vdots & a_{ij} & \vdots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,x-1} & a_{n,x} \end{bmatrix} \tag{19}$$

where, a denotes matrix of vectors position, (a_{ij}) represents the positions vectors, n and x represent gazelles number and optimization search space.

Fitness Function

$$Fitness\ Function = \min\left(|< R \mid L >|^2\right) \tag{20}$$

where, $|< R \mid L >|^2$ represent the Minimizing the kernel which improves performance by reducing error and enhancing the alignment of quantum states, leading to better detection of attack patterns.

Exploitation Phase

During grazing, it is hypothesized that the gazelles move in a Brownian motion pattern is shown in (21),

$$gazelle_{j+1} - gazelle_j + m \cdot L * L_c * \cdot \left(Elite_j - L_c * \cdot gazelle_j\right) \tag{21}$$

where, $gazelle_{j+1}$ represents next iteration solution, $gazelle_j$ represents current iteration solution, m denotes the gazelles grazing speed, L_c denotes Random integers in a vector that mimics Brownian motion and L represents uniform random numbers of vectors ranging from 0 to 1.

Exploration Phase

Gazelle's behavior of spotting predator is presented in (22),

$$\overrightarrow{gazelle}_{j+1} - \overrightarrow{gazelle}_j + B.\mu.\vec{L} * \cdot \vec{L_e} * \cdot \left(\overrightarrow{Elite}_j - \vec{L_e} * \cdot \overrightarrow{gazelle}_j\right)$$

(22)

where, B represents speed that gazelle achieve, $\vec{L_e}$ represents vector of Levy distributions random numbers, Behavior of the predator chasing the gazelle is shown in (23),

$$\overrightarrow{gazelle}_{j+1} - \overrightarrow{gazelle}_j + B \cdot \mu \cdot EP * \cdot \vec{L_c} * \cdot \overrightarrow{Elite}_j - \vec{L_e} + \cdot \overrightarrow{gazelle}_j \tag{23}$$

where, EP represents predator cumulative effect, QRP_s modelling shown in (24),

$$\overrightarrow{gazelle}_{j+1} = \begin{cases} \overrightarrow{gazelle}_j + DE\left[\overrightarrow{MS} + \vec{L} * \cdot \left(\overrightarrow{SJ} - \overrightarrow{TJ}\right)\right] * \cdot \vec{S}; ifo \le QRP_s \\ \overrightarrow{gazelle}_j + \left[QRP_s(1-o) + o\right]\left(\overrightarrow{gazelle}_{o1} - \overrightarrow{gazelle}_{o2}\right), else \end{cases}$$

(24)

where, (QRP_s) represents the predator success rate , $\vec{S}$ represents the random number of binary vector r on interval [0,1].

3.2.4. X-IDS (Explainable Intrusion Detection System) with SHAP

QKCGS models, while highly effective, are often difficult to understand due to their black-box nature. This opacity complicates decision-making and interpretation, making it challenging to trust their outcomes, especially in IDS. To enhance trust and transparency, integrating Explainable AI methods like SHAP can provide clear insights into feature contributions, improving the reliability of IDS for accurate attack classification. It evaluates the model's impact on each feature's output by calculating Shapley values, which represent each prediction of feature contribution. This helps in identifying both positive and negative

influences of different features on attack detection, making the model's behavior more interpretable. In the single Instance the $SHAP$ can be calculated in (25).

$$H(t) = V_0 + \sum_{j=0}^{O} V_j t_j \tag{25}$$

where t represents the new features, O denotes maximum size, and V_j denotes shapely value. By analyzing the average Shapley values across data samples, the method highlights the key features that significantly influence the model classification of attacks. This feature importance analysis helps in refining the IDS, making it more transparent and reliable for real-time attack detection. The important features selected can be shown in (26).

$$\mathrm{Im}\,p\ Gt_k = \sum_{j=1}^{o} \| V_k(Y_j) \| \tag{26}$$

where, o represents number of data samples and Shapely value of j th feature input by $\mathrm{Im}\,p\ Gt_k$. Flow Chart for Proposed Methodology is given in Figure 2.

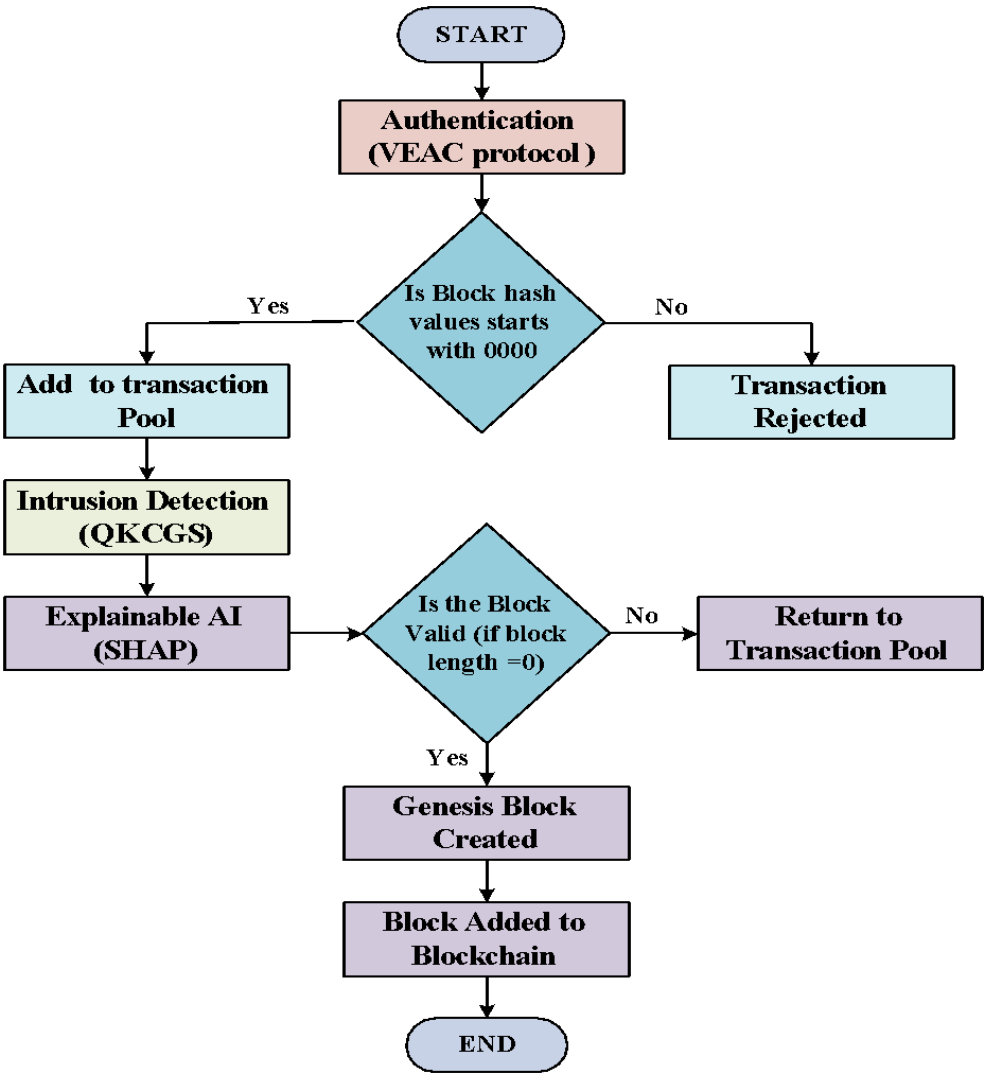


Figure 2: Flow Chart for Proposed Methodology

The proposed VEAC protocol enhances secure, anonymous communication across IoT, fog and cloud layers using elliptic curve signatures and blockchain technology for key generation and verification. SCO optimizes the computational efficiency of authentication by reducing overhead. QKCGS improves intrusion detection through a quantum kernel self-attention mechanism, capturing both short- and long-term attack patterns. GOA is employed to refine the detection model. The system is further enhanced with SHAP for explainable AI, providing insights into accurate attack classification. Together, these methods ensure efficient, secure communication and robust intrusion detection in IoT environments.

4. Experimental Results and Discussion

The system improves security by using blockchain-based authentication for immutable and transparent credential storage. It enhances intrusion detection with the QKCGS, which effectively processes sequential data. The system is tested and configured on a Windows 10 environment, equipped with 32GB of RAM and running Python 3.12.7 for efficient processing. Development and debugging are carried out using Visual Studio Code, which provides a streamlined coding environment. Machine learning experiments are set up with 10 epochs and 100 iterations to ensure optimal performance and reliability. It has also been benchmarked against existing methods, including the DBN [18], Convolutional Neural Networks (CNN) [19], CPS [20], LSTM [22], BNN [23], and WSN [24], to assess its performance. The results of the existing techniques are referred to the reference.

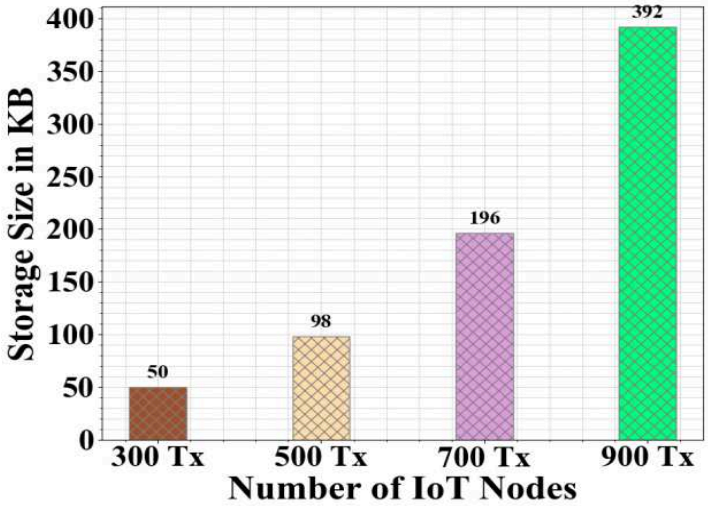


Figure 3: Performance Analysis of Blockchain Authentication for Transaction Storage Size

Figure 3 presents the performance analysis of blockchain authentication for the transaction storage required for blockchain authentication, showing the relationship between storage size (in KB) and the number of IoT nodes. With 300 transactions, the storage reaches 50 KB; with 500 transactions, it reaches 98 KB; with 700 transactions, it reaches 196 KB; and with 900 transactions, the storage requirement increases to 392 KB.

4.1. Dataset Descriptive Analysis

The CIC-IDS-2017 [30] dataset is a comprehensive collection of network traffic data, designed for the detection and analysis of cyber intrusions. It contains over 80 features, including time-based and flow-based attributes, and is designed to represent different types of network attacks. CSE-CIC-IDS2018 [31] dataset, also referred to as IDS 2018, builds upon its predecessor by offering enhanced granularity and variety in attack types. It contains

labeled data with more than 80 features capturing network traffic, including flow characteristics, packet-level information, and application-layer protocols.

Table 3: Distribution of Attacks in CIC-IDS-2017 Dataset

Attack Type	Samples	Percentage (%)
BENIGN	2096484	77.79
Dos Hulk	172849	6.41
Comb	172800	6.41
DDoS	128016	4.75
Port Scan	90819	0.38
DoS Golden Eye	10293	0.25
FTP-Patator	5933	0.22
Dos Slowloris	5385	0.20
DoS Slowhttptest	5228	0.19
SSH-Patator	3219	0.12
Bot	1953	0.07
Web Attack Brute Force	1478	0.05
Web Attack XSS	652	0.02
Infiltration	36	0.00
Web Attack SQL Injection	21	0.00
Hearthleed	11	0.00

Table 3 shows the distribution of CIC-IDS-2017 dataset with 16 attack classes, with a majority of benign samples: 2,096,484 samples, representing 77.79%. The dataset also includes Denial of Service (DoS) attacks like DoS Hulk with 172,849 samples (0.41%), Comb with 172,800 samples (6.41%), DoS Golden Eye with 10,293 samples (0.25%), and DDoS with 128,016 samples (4.75%). Other attack types include Port Scan with 90,819 samples (3.37%), DoS Golden Eye with 10,286 samples (0.38%), and various DoS subtypes such as Slowloris with 5,385 samples (0.20%) and Slowhttptest with 5,228 samples (0.19%). Additional attacks include FIP-Patator with 5,933 samples (0.22%), SSH-Patator with 3,219 samples (0.12%), Bot with 1,953 samples (0.07%), and Web Attacks like Brute Force with 1,470 samples (0.05%) and XSS with 652 samples (0.02%). The dataset also contains rare

attacks such as infiltration with 36 samples (0.00%), SQL injection with 21 samples (0.00%), and Heartbleed with 11 samples (0.00%).

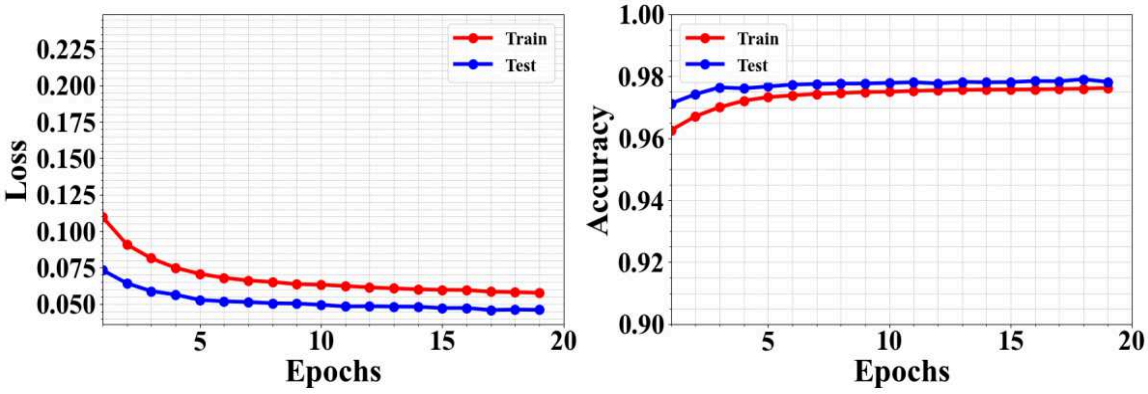


Figure 4: Loss vs. Accuracy computed with VEAQS method on CIC-IDS-2017 Dataset

Figure 4 illustrates the loss and accuracy metrics computed using the VEAQS method on the CIC-IDS-2017 dataset, with both training and testing conducted over 20 epochs. The loss for the training data is 0.115, while for the testing data, it is 0.075. In terms of accuracy, the model achieves a training accuracy of 0.96 and a testing accuracy of 0.97. These results indicate strong performance with minimal loss and high accuracy across both training and testing datasets.

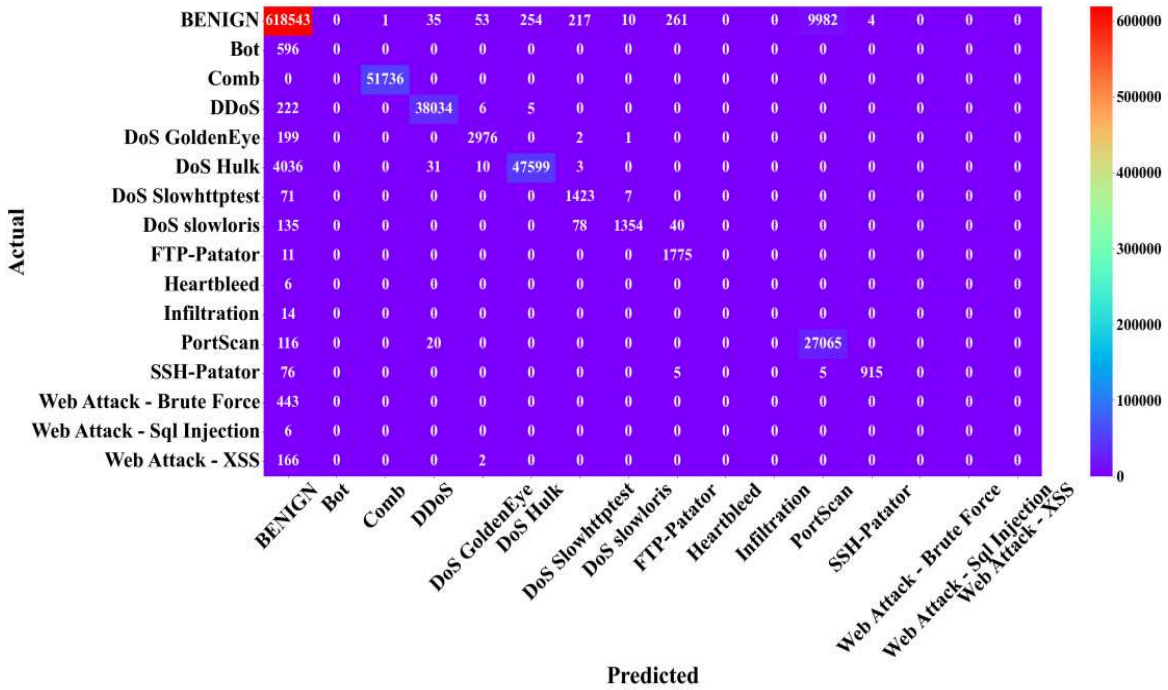


Figure 5: Confusion matrix of CIC-IDS-2017 Dataset

Figure 5 presents the confusion matrix for the CIC-IDS-2017 dataset, which includes 16 attack categories: BENIGN, DoS Hulk, Comb, DDoS, Port Scan, DoS Golden Eye,

FIP-Patator, DoS Slowloris, DoS Slowhttptest, SSH-Patator, Bot, Web Attack Brute Force, Web Attack XSS, Infiltration, Web Attack Sql Injection, and Heartbleed. Among these, the BENIGN class has the highest value with 618,543, indicating a large number of benign traffic instances. The second largest value is found in the COMB class, with 51,736 instances. The values for all other classes range from 0 to 600,000. This matrix reflects the performance of the model in classifying both normal and attack traffic across different categories.

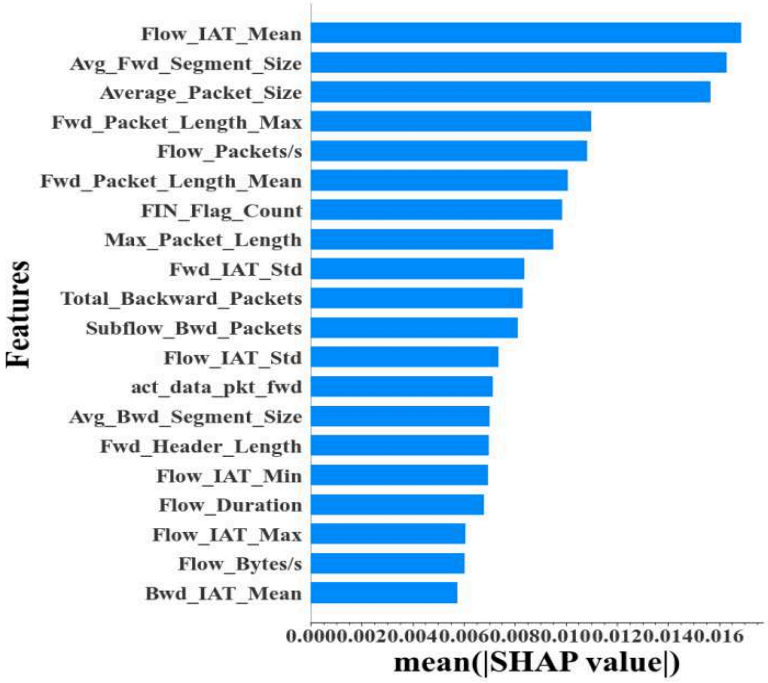


Figure 6: SHAP values for CIC-IDS- 2017 dataset using summary plot

Figure 6 presents the SHAP values for the CIC-IDS-2017 dataset using a summary plot. The dataset includes features such as Flow_IAT Mean, Avg Fwd Segment Size, Average Packet Size, Fwd_Packet_Length_Max, Flow Packets/s, Fwd_Packet_Length_Mean, FIN Flag Count, Max_Packet_Length, Fwd_IAT_Std, Total Backward Packets, Subflow Bwd Packets, Flow_LAT_Std, act_data_pkt_fwd, Avg Bwd Segment Size, Fwd_Header_Length, Flow_IA_Min, Flow Duration, Flow_IAT_Max, Flow_Bytes/s, and Bwd_LAT_Mean. The mean SHAP values for these features range from 0.000 to 0.016. Among these, the Flow_IAT Mean feature has the highest mean SHAP value of 0.016, while the Bwd_LAT_Mean feature has the lowest mean value at 0.005. The remaining features have mean SHAP values that lie between these two extremes. These values provide insights into the relative importance of each feature in the model's predictions. Overall, the summary plot helps in understanding the contribution of individual features to the model's output.

Table 4: Distribution of Attacks in CSE-CIC-IDS2018 Dataset

Attack Type	Samples	Percentage (%)
Benign	10732	18.05
Bot	9376	15.77
DDOS attack-HOIC	9300	15.64
DDoS attacks-LOIC-HTTP	8388	14.11
DoS attacks-Golden Eye	7250	12.19
DoS attacks-Hulk	6850	11.52
DoS attacks-Slow HTTP Test	1250	2.10
DoS attacks-Slowloris	1110	1.87
FTP-Brute Force	1350	2.27
Infiltration	1850	3.11
SSH-Brute Force	2000	3.36

Table 4 shows the CSE-CIC-IDS2018 dataset includes various types of attacks, each with its own distribution of samples. The largest proportion of data is benign traffic, accounting for 18.05% (10,732 samples). The next most common attack types are Bot attacks at 15.77% (9,376 samples) and DDoS-HOIC at 15.64% (9,300 samples). Other notable attacks include DDoS-LOIC-HTTP (14.11%, 8,388 samples) and DoS-Golden Eye (12.19%, 7,250 samples). Smaller percentages include DoS-Hulk (11.52%, 6,850 samples), DoS-Slow HTTP Test (2.10%, 1,250 samples), and DoS-Slowloris (1.87%, 1,110 samples). Other attacks like FTP-Brute Force, Infiltration, and SSH-Brute Force make up smaller but significant portions of the dataset at 2.27%, 3.11%, and 3.36%, respectively.

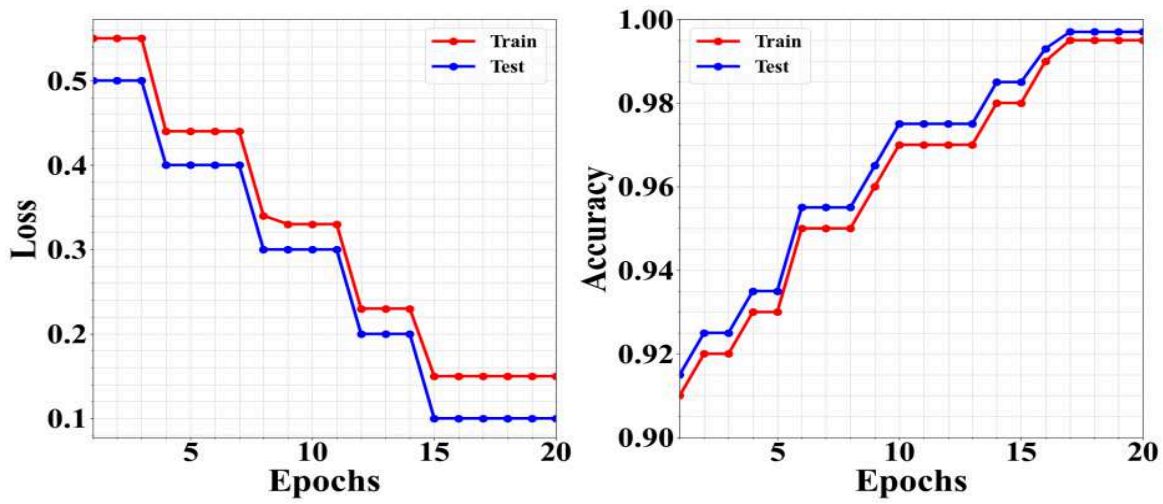


Figure 7: Loss vs. Accuracy computed with VEAQS method on CSE-CIC-IDS2018 Dataset

Figure 7 shows the loss and accuracy metrics obtained using the VEAQS method on the CSE-CIC-IDS2018 dataset, with both training and testing performed over 20 epochs. The training loss is 0.55, while the testing loss is 0.5. The model achieves a training accuracy of 0.992 and a testing accuracy of 0.998. These results indicate strong performance, with minimal loss and near-perfect accuracy on both the training and testing datasets.

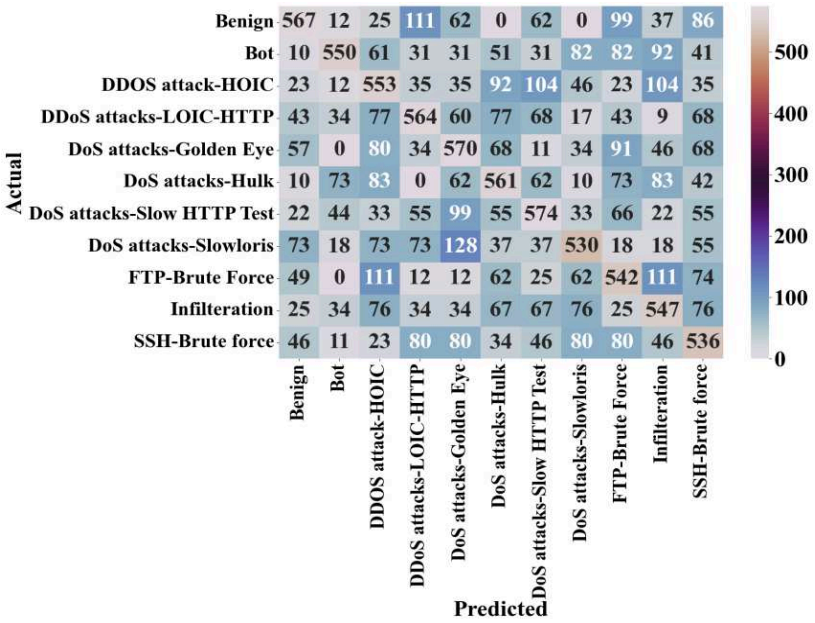


Figure 8: Confusion matrix of CSE-CIC-IDS2018 Dataset

Figure 8 displays the confusion matrix for the CSE-CIC-IDS2018 dataset, which consists of 11 classes: Benign, Bot, DDOS attack-HOIC, DoS attacks-Golden Eye, DDoS attacks-LOIC-HTTP, DoS attacks-Hulk, DoS attacks-Slow HTTP Test, DoS attacks-Slowloris, FTP-Brute Force, infiltration and SSH-Brute Force. The values in the matrix range from 0 to 600, with some classes having significantly higher values. The "DoS attacks-Golden Eye" class has a value of 570, and the "Benign" class has a value of 567. Other classes show values ranging between 0 and 564. These values represent the model's performance across the various attack and normal traffic categories.

4.2. Comparative Analysis

Table 5: Comparison of Time analysis of Block creation with

Existing and proposed

IOT Nodes	Transaction time in (ms)							
	300 Tx		500 Tx		700Tx		900 Tx	
	Existing	Proposed	Existing	Proposed	Existing	Proposed	Existing	Proposed
10	24.14	20.23	26.23	22.32	31.38	26.43	35.34	30.65
20	35.34	28.28	41.42	29.34	41.42	30.54	46.77	32.43
30	36.21	29.42	39.35	30.55	41.5	32.28	47.91	35.87
40	40.24	30.12	51.42	38.23	49.65	40.22	54.92	42.45
50	41.29	31.50	49.59	40.76	67.78	42.90	72.97	45.23

Table 5 compares the transaction time for block creation between existing [27] and proposed systems across different IoT node counts and transaction volumes. The proposed system consistently shows lower transaction times compared to the existing system, indicating improved efficiency.

Table 6: Overall Comparison table existing and proposed methodology for CIC-IDS- 2017 Dataset

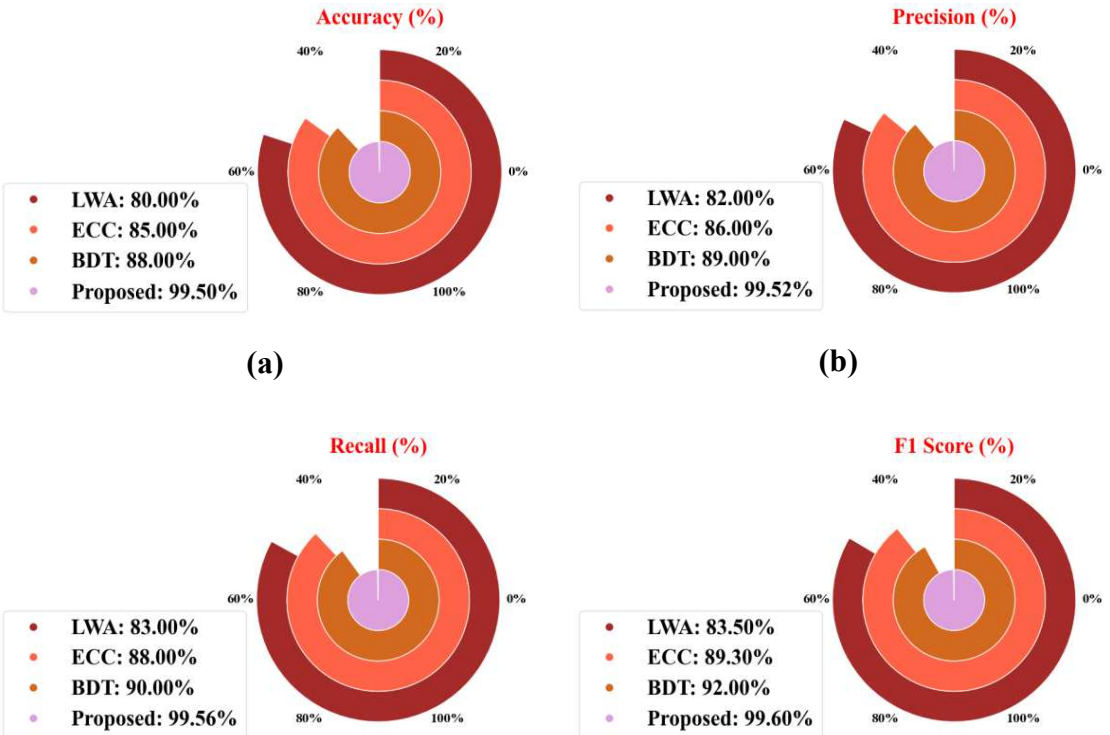
Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DNN [27]	93.85	93.21	94.88	93.45
BiGRU [22]	95.43	94.72	96.23	95.77
BiGRNN [20]	97.12	97.86	97.32	97.34
VEAQS(Proposed)	99.79	99.42	98.96	98.98

Table 6 presents a comparison between the existing methodologies and the proposed VEAQS approach using the CIC-IDS-2017 dataset across four performance metrics. The Deep Neural Network (DNN) method achieves an accuracy of 93.85%, precision, recall, and F1-score values of 93.21%, 94.88%, and 93.45%, respectively. The BiGRU model shows an improvement of 95.43%, 94.72%, 96.23% and 95.77%. The proposed VEAQS method outperforms all existing methods, achieving the highest accuracy at 99.79%, precision at 99.42%, recall at 98.96%, and F1-score at 98.98% respectively.

Table 7: Overall Comparison table existing and proposed methodology for CSE-CIC-IDS2018 Dataset

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN+LSTM[32]	92.51	93.49	83.12	91.61
LSTM [33]	93.87	95.82	91.23	94.28
DNN [34]	95.91	98.23	96.72	96.45
VEAQS(Proposed)	99.19	99.42	98.85	99.01

The comparison table 7 presents the performance of various methodologies applied to the CSE-CIC-IDS2018 dataset. The CNN+LSTM model achieves an accuracy of 92.51%, precision of 93.49%, recall of 83.12%, and an F1-score of 91.61%. The LSTM model performs slightly better with 93.87% accuracy, 95.82% precision, 91.23% recall, and an F1-score of 94.28%. The DNN model shows the highest performance among the existing methods with 95.91% accuracy, 98.23% precision, 96.72% recall and 96.45% F1-score. The proposed methodology outperforms all previous models, achieving 99.19% accuracy, 99.42% precision, 98.85% recall and 99.01% F1-score, demonstrating a significant improvement in all metrics.



(c) **(d)**

Figure 9: Comparison of existing and proposed model for Authentication

Figure 9 presents a comparison of the existing authentication methods like LWA [18], ECC [24] and BDT [22] with the proposed VEAC protocol across various metrics. For accuracy, the existing methods show LWA at 80%, ECC at 85% and BDT at 88%, while the proposed protocol achieves a significant improvement with 99.50%. In terms of precision, LWA, ECC and BDT have values of 82%, 86%, and 89%, respectively, whereas the VEAC protocol reaches 99.52%. For recall, the existing methods score 83% for LWA, 88% for ECC, and 90% for BDT, while the VEAC protocol outperforms them with a recall of 99.56%. Finally, for the f1-score, LWA achieves 83.50%, ECC 89.30%, and BDT 92%, whereas the VEAC protocol leads with an f1-score of 99.60%. These results demonstrate the substantial improvements in performance offered by the proposed VEAC protocol compared to the existing models.

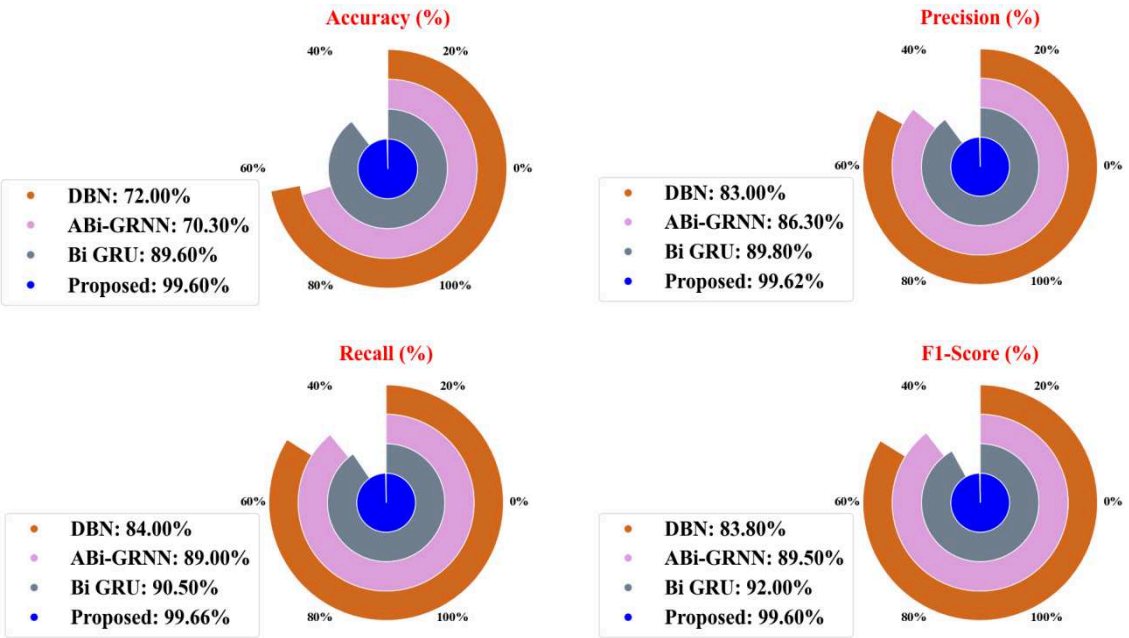


Figure 10: Comparison of existing and proposed model for Intrusion Detection

Figure 10 compares the performance of existing models like DBN [18], ABi-GRNN [20], Bi GRU [22], and the proposed QKCGS model for intrusion detection across four metrics. The DNN model shows the lowest performance with 72% Accuracy, while ABi-GRNN improves to 70.3% accuracy and higher precision and Recall. The Bi GRU model performs better with 89.6% accuracy, 89.8% precision and 90.5% recall. However, the proposed QKCGS model significantly outperforms all others, achieving 99.6% Accuracy, 99.62%

precision, and 99.66% recall and 99.6% f1-score, highlighting its superiority in intrusion detection.

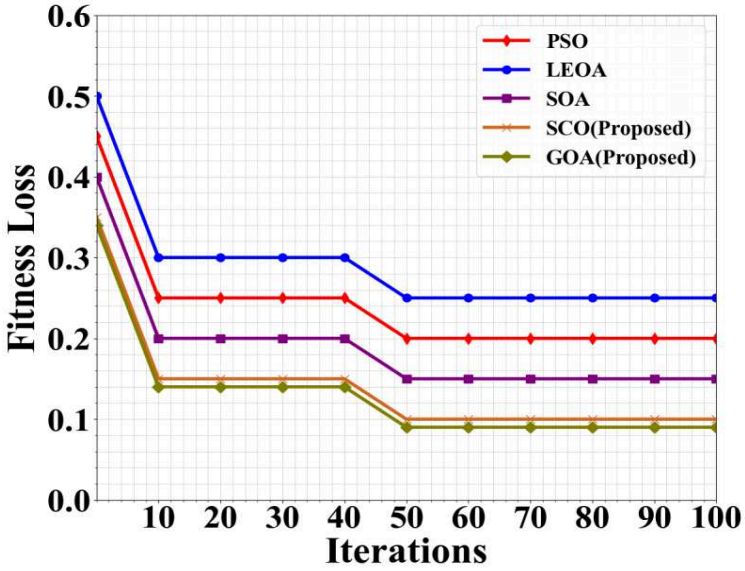


Figure 11: Fitness function

Figure 11 shows the fitness function performance of various optimization methods, including PSO, LEOA, SOA, as well as the proposed methods SCO and GOA, with iteration values up to 100. The loss values at the final iteration are 0.45 for PSO, 0.5 for LEOA, 0.4 for SOA, 0.35 for SCO, and 0.34 for GOA. These results demonstrate that the proposed methods, SCO and GOA, achieve lower loss values, indicating better performance and higher accuracy compared to the existing optimization techniques.

4.3. Cross Validation

Table 8: Cross Validation

Number of Folds	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Fold 2	89.5	85.37	82.55	90.97
Fold 4	92.32	89.35	87.82	93.83
Fold 6	93.21	93.45	93.91	95.98
Fold 8	97.93	96.54	95.92	98.86
Fold 10	98.86	98.19	97.72	99.12

Table 8, presents the results of cross-validation for a model evaluated across different numbers of folds. As the number of folds increases, the model performance also improves. For Fold 2, the accuracy is 89.5%, with precision at 85.37%, recall at 82.55%, and an F1

score of 90.97%. For Fold 4, accuracy rises to 92.32%, precision to 89.35%, recall to 87.82%, and F1 score to 93.83%. In Fold 6, the accuracy further improves to 93.21%, with precision at 93.45%, recall at 93.91%, and an F1 score of 95.98%. Fold 8 shows significant gains, achieving 97.93% accuracy, 96.54% precision, 95.92% recall, and an F1 score of 98.86%. Finally, Fold 10 yields the highest results, with accuracy at 98.86%, precision at 98.19%, recall at 97.72%, and an F1 score of 99.12%. This trend indicates that as the number of folds increases, the model becomes more reliable and accurate.

5. Conclusion

In conclusion, the proposed framework effectively integrates blockchain-based authentication, quantum-enhanced intrusion detection, and explainable AI to deliver a robust security solution for IoT systems. By leveraging decentralized ledger technology, it ensures data integrity, while the QKCGS enhances the accuracy and efficiency of attack detection. PUF has been used for improving authentication without storing the keys. The combination of deep learning and blockchain reduces computational overhead, optimizes key generation, and improves real-time response to threats. Future work will focus on further optimizing the quantum models for higher scalability, integrating advanced privacy-preserving techniques for user data, and exploring additional attack detection capabilities in dynamic IoT environments. Enhancing the real-time performance of the system through hardware acceleration and expanding the use of Shapley explanations for more complex attack scenarios will be key areas of development. Testing and deployment in large-scale IoT networks will validate the framework's effectiveness in diverse operational settings.

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Disclosure of interest

No potential competing interest was reported by the authors

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Compliance with Ethical Standards

1. **Potential Conflict of Interest Disclosure:**

   None of the authors have any potential conflicts of interest.

2. **Declaration on Human and Animal Rights**

a. Ethical Approval: All relevant national and/or academic regulation for the use and care of animals was strictly adhered to.

3. **Informed Consent**

   Formal consent is not essential for this particular type of research.

Replication of results

No results are presented

Data availability statements

As no datasets were created or examined for this research, data sharing is not appropriate.

Authors' contribution

Every author has made an equal contribution to the work. All authors reviewed the manuscript.

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